Midterm Exam

- Do not open this quiz booklet until directed to do so. Read all the instructions on this page.
- When the quiz begins, write your name on the top of every page of this quiz booklet.
- You have 80 minutes to earn a maximum of 80 points. Do not spend too much time on any one problem. Skim them all first, and attack them in the order that allows you to make the most progress.
- **You are allowed three double-sided letter-sized sheet with your own notes**. No calculators, laptops, cell phones, or other programmable or communication devices are permitted.
- Write your solutions in the space provided. Pages will be scanned and separated for grading. If you need more space, write "Continued on S1" (or S2, S3) and continue your solution on the referenced scratch page at the end of the exam.
- Do not waste time and paper rederiving facts that we have studied in lecture, recitation, or problem sets. Simply cite them.
- **Pay close attention to the instructions for each problem**. Depending on the problem, partial credit may be awarded for incomplete answers.

Problem	Parts	Points
0: Information	2	2
1: True or False?	7	28
2: Does It Permute?	1	15
3: Fiesty Ciphers	3	20
4: Combiners!	1	15
Total		80

Name: _____

School Email: _____

Problem 1. [28 points] **True or False?** (7 parts)

For each of the following questions, tick either **T** (True) or **F** (False) box. Each problem is worth 4 points. **You will get 4 points for a correct answer, 0 points for an incorrect answer, and 2 points if you leave it blank.** No justification is necessary, and no partial credit will be given.

(a) [**T**] [**F**] The function $\mu(n) := 1/2^{\sqrt{\log_2 n}}$ is negligible.

(b) [**T**] [**F**] For any one-way permutation $f(x)$, $f(f(x))$ is also a one-way permutation.

(c) [**T**] [**F**] Any length-doubling pseudo-random generator (PRG) family is also a one-way function (OWF) family.

(d) [**T**] [**F**] There is a known proof that public-key encryption is possible assuming only that one-way functions (OWF) exist.

(e) **T** **F** Assume that one-way permutations exist. Then, there exists a one-way permutation $f : \{0,1\}^n \to \{0,1\}^n$ such that $f(f(x)) = x$ for all $x \in \{0,1\}^n$.

(f) **T** **F** There exists a function $h : \{0,1\}^n \to \{0,1\}$ such that for all one-way fuctions $f : \{0,1\}^n \to \{0,1\}^n$, h is a hardcore predicate for f.

(g) **T** **F** The following two distributions are computationally indistinguishable:

$$(g, g^x, g^y, g^{xy}) \qquad \text{and} \qquad (g, g^x, g^y, g^z)$$

where g is uniformly drawn generator of the group $\mathbb{Z}_p^*$, and x, y and z are uniformly drawn from the set $\{1, 2, \ldots, p-1\}$.

Problem 2. [15 points] **Does It Permute?**

Let $f : \{0,1\}^n \to \{0,1\}^n$ be a one-way permutation. Construct a one-way permutation $g : \{0,1\}^n \to \{0,1\}^n$ such that $g(0^n) = 0^n$. (Here, 0^n refers to the n-bit string of zeroes.) Prove that your construction is a permutation and that it is one-way.

Problem 3. [20 points] **Fiesty Ciphers** (3 parts)

Let $\mathcal{F} = \{F_k : \{0,1\}^n \to \{0,1\}^n : k \in \{0,1\}^n\}$ be a pseudorandom function (PRF) family. We say that $\mathcal{G} = \{G_k : \{0,1\}^n \to \{0,1\}^n : k \in \{0,1\}^m\}$, for $m = \mathsf{poly}(n)$, is a pseudorandom *permutation* (PRP) family if it is a PRF family, and additionally, (1) each G_k is a permutation; and (2) each G_k is invertible in polynomial time given k. The security definition for a PRP is exactly the same as that for a PRF.

(a) [4 points] Define $G_k : \{0,1\}^{2n} \to \{0,1\}^{2n}$ to be

$$G_k(x\|y) := y\|(F_k(y) \oplus x) \; .$$

Show that it is a permutation by writing down the inverse function $G_k^{-1}(\cdot\|\cdot)$ explicitly (Note: We are **not yet** asking you to show that this is a *pseudorandom* permutation.)

(b) [4 points] Show that G_k is **not** a PRP family.

(c) [12 points] Now, let's try doing this transformation one more time. Define

$$H_{k_1,k_2}(x\|y) := G_{k_2}(G_{k_1}(x\|y)) = \big(F_{k_1}(y) \oplus x\big)\|\big(F_{k_2}(F_{k_1}(y) \oplus x) \oplus y\big).$$

Prove that H_{k_1,k_2} is **not** a PRP family.

Problem 4. [15 points]

Noga and Zoe are participants in an encryption design competition. They each came up with a novel public-key bit encryption scheme: let's call Noga's scheme $\mathcal{E}_N = (\mathsf{Gen}_N, \mathsf{Enc}_N, \mathsf{Dec}_N)$ and Zoe's scheme $\mathcal{E}_Z = (\mathsf{Gen}_Z, \mathsf{Enc}_Z, \mathsf{Dec}_Z)$. Shafi, the judge, inspected both schemes and figured out that one of them is IND-CPA secure but the other one isn't. Unfortunately, she disappeared before telling them which one is secure!

Noga and Zoe decided to join forces and submit the following scheme $\mathcal{E} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ to the competition. Your task is to use a hybrid argument to prove that if either $\mathcal{E}_N$ or $\mathcal{E}_Z$ is IND-CPA secure, so is $\mathcal{E}$. State your hybrids clearly.

- $\mathsf{Gen}(1^\lambda)$: Generate keys $(pk_N, sk_N) \leftarrow \mathsf{Gen}_N(1^\lambda)$ and $(pk_Z, sk_Z) \leftarrow \mathsf{Gen}_Z(1^\lambda)$. Output

$$pk = (pk_N, pk_Z) \qquad \text{and} \qquad sk = (sk_N, sk_Z)\ .$$

- $\mathsf{Enc}(pk, b)$ where b is a bit: generate a random bit $r \leftarrow \{0,1\}$ and output $c = (c_N, c_Z)$ where

$$c_N \leftarrow \mathsf{Enc}_N(pk_N, r) \qquad \text{and} \qquad c_Z \leftarrow \mathsf{Enc}_Z(pk_Z, r \oplus b)$$

- $\mathsf{Dec}(sk, c)$: let $b_N \leftarrow \mathsf{Dec}_N(c_N)$ and $b_Z \leftarrow \mathsf{Dec}_Z(c_Z)$. Output $b_N \oplus b_Z$.

Note: Make sure you handle both cases in your proof.